BUILD2LAUNCH · NICKHI CAMPUS AMBASSADOR

Vibecoding & AI Workshop

Turn an idea into a working AI-powered application — even with minimal coding experience.

2 Hours | Interactive + Live Demo | Idea → Deploy

WORKSHOP GOAL

What you will leave with

Mindset

- Startups begin with problems, not code
- AI is a teammate, not magic
- Ship version one, then improve

Skills

- Write better prompts
- Generate and run app code
- Use datasets + ML + Gemini together
- Deploy a live app

> Today’s promise: Idea → Prompt → Code → Dataset → ML Model → Gemini → Deploy

AGENDA · 120 MINUTES

Training schedule

No.	Session	Focus	Time
1	Startups, AI & Vibecoding	Concepts	10 min
2	AI Models & Tools	Gemini, VS Code, GitHub, Kaggle	10 min
3	Prompt Engineering	Role + Goal + Constraints	10 min
4	Live Demonstration	Build ReviewRadar	80 min
5	Closing & Next Steps	Practice plan	10 min

SESSION 1 · TEACHING

What is a startup, really?

1. Problem

Find a real pain point people feel often.

2. Solution

Design the simplest useful product that helps.

3. Ship

Build a small version, test with users, improve.

> Example problem today: customer reviews are emotional and messy. Businesses need fast understanding.

SESSION 1 · TEACHING

What is Artificial Intelligence?

In simple words

AI systems learn patterns from data and help with tasks like writing, classifying, recommending, and generating code.

Useful types for builders

- **Generative AI** — writes text, code, explanations
- **Machine Learning** — predicts labels/scores from examples

> In this workshop we use both: an ML classifier + Gemini generative AI.

SESSION 1 · TEACHING

What is vibecoding?

Building software by describing what you want in plain language, generating code with an LLM, then testing and improving quickly.

Describe idea → Write prompt → Generate code → Run & fix → Ship

You still need

- Clear thinking
- Testing
- Basic debugging

AI helps with

- Boilerplate code
- Faster iteration
- Explanations & rewrites

SESSION 2 · TOOLS

Your workshop toolkit

Gemini Flash-Lite

Fast generative AI model for explanations and replies via API.

VS Code

Where code lives, edits happen, and the app runs locally.

GitHub

Stores your project online and connects to deployment.

Kaggle

Source of real datasets for training a small ML model.

Also: ChatGPT / Claude / DeepSeek for code generation

Vercel for deployment

SESSION 2 · CRITICAL CONCEPT

Gemini ≠ Kaggle training

Gemini Flash-Lite

- Generative AI
- Accessed with an API key
- Great at writing & explaining
- Not trained on your CSV through the API

Separate ML model

- Trained on Kaggle-style dataset
- Learns Positive / Negative patterns
- Exports to **model.json**
- Used for prediction inside the app

> Final product = ML prediction + Gemini explanation/reply.

SESSION 3 · PROMPT ENGINEERING

The 3-part prompt formula

1. Role

Who should the AI act as?

“Act as a senior Next.js engineer…”

2. Goal

What exact outcome do you want?

“Build ReviewRadar with Analyze button…”

3. Constraints

Stack, style, limits, format.

“TypeScript, Tailwind, one page UI…”

> Prompt quality matters more than which chatbot logo you use.

SESSION 3 · PRACTICE

Bad prompt vs good prompt

WEAK

Bad

```
Make a review app
```

STRONG

Good

```
Act as a senior Next.js engineer.
Build ReviewRadar: one text box, Analyze button,
sentiment result, Gemini explanation + suggested
reply.
Use App Router, TypeScript, Tailwind.
Provide full file contents and run commands.
```

SESSION 3 · TEACHING

Which coding LLM should you use?

Model	Strength in workshops
ChatGPT (GPT)	Great general coding partner and fast iteration
Claude	Strong with longer code and careful edits
DeepSeek	Strong coding model and useful alternative

> Use whichever you have access to. The prompt structure stays the same.

SESSION 4 · LIVE DEMO

Today we build: ReviewRadar

A feedback analyzer for startups and campus projects.

Paste review → ML predicts sentiment → Gemini explains → Suggested reply → Deploy

Why this demo?

- Clear business story
- Uses dataset + generative AI
- Beginner-friendly UI

Stack

- Next.js + TypeScript
- scikit-learn ML model
- Gemini Flash-Lite
- Vercel

SESSION 4 · DEMO ROADMAP

8 steps we will demonstrate

1. **API key** — Google AI Studio for Gemini
2. **Idea → prompt** — convert the product idea into instructions
3. **Code generation** — GPT / Claude / DeepSeek
4. **VS Code setup** — paste, install, run locally
5. **Dataset** — small labeled CSV from Kaggle-style data
6. **Train ML model** — Logistic Regression → model.json
7. **Integrate Gemini** — explanation + polite reply
8. **Test & deploy** — verify flow, ship on Vercel

TEACHING CHECKPOINT

How the system works

Input

User pastes a customer/product review into the web app.

ML Model

Classifies text as Positive or Negative using patterns learned from dataset examples.

Gemini

Reads the review + prediction, then writes an explanation and suggested reply.

> Participants should be able to explain this architecture in one sentence before we deploy.

TEACHING · DATASET

What makes a good workshop dataset?

Keep it small & clear

- Text + label columns
- About 2,000–5,000 rows is enough
- Labels like 0/1 or negative/positive

What we teach with it

- Clean / prepare data
- Train a simple classifier
- Export model for the app

```
text,label
"I love this product",1
"Terrible quality, never again",0
```

TEACHING · ML + AI TOGETHER

Two engines, one product

Part	Job in ReviewRadar	Output
ML classifier	Detect sentiment from learned patterns	Positive / Negative + confidence
Gemini Flash-Lite	Generate human-friendly response	Explanation + suggested reply
Web app	Connect both and show results	Usable product UI
Vercel	Make it public	Live URL

LIVE DEMO SUPPORT

Test cases to try on stage

Positive

```
I absolutely love this product. Fast delivery and amazing quality!
```

Negative

```
Terrible experience. Broke on day one and support ignored me.
```

Mixed

```
Good idea overall, but setup was confusing and a bit slow.
```

> Mixed/sarcastic examples teach that AI is useful, but not perfect.

SESSION 5 · RECAP

What we covered

Concepts

- Startups = problem → solution → ship
- Vibecoding workflow
- Prompt formula: Role + Goal + Constraints

Build path

- LLM generates app code
- Dataset trains separate ML model
- Gemini powers generative features
- Vercel deploys the product

SESSION 5 · NEXT STEPS

Your 7-day practice plan

1. **Day 1–2:** Rebuild ReviewRadar locally
2. **Day 3:** Change the idea (spam detector, campus complaint classifier, interview coach)
3. **Day 4:** Improve prompts and UI
4. **Day 5:** Deploy to Vercel
5. **Day 6–7:** Share with a friend and collect feedback

> A PDF handout with workflow, tools, and prompts will be shared for independent practice.

REMEMBER

You don’t need to know everything.

You need a problem, a clear prompt, and the courage to ship version one.

Problem → Prompt → Prototype → Deploy

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Thank you

Let’s turn ideas into working AI-powered applications.

NICKHI Campus Ambassador

Questions welcome

Now go build